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Learn About Hard Disk Drives

UPGRADING

A Z-248 TO A 386SX

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Background

I originally purchased my 8 MHz Z-248 in April of 1987. The package included a 80286 processor, 512 KB of RAM, one 5-1/4" floppy (360 KB), one Seagate 20 MB (model 225) hard disk drive, one serial port, one parallel port, an EGA adapter (256 KB), an 84-key keyboard, ten 5-1/4" diskettes, MS-DOS 3.2, GW-BASIC, Windows 1.0, diagnostic software, an EGA color monitor (model 1380), and 10 slots (four already filled with a disk controller card, a CPU card, an I/O card, and a video card: leaving 6 open slots).

Minor Upgrades

Through the years I made minor hardware upgrades to accommodate my purchases of the latest software. I purchased a serial Logitech mouse and had to install a serial card since my original serial port was being used by the modem. I also purchased 128 KB of memory to fill the memory from 512 KB to 640 KB.

Having to deal with the new smaller floppies and my running out of hard disk space, I purchased a Teac high density (1.4 MB) 3-1/2" disk drive and a 40 MB Seagate (model 251-1) hard drive. I purchased updated Zenith Data Systems ROMs to access the 1.4 MB drive. Zenith Data Systems, thank their technical expertise, not only provided a disk controller card that allowed two floppy drives, and two hard drives; but also allowed the space to fit each internal bay of the machine with required power wires from the 200 watt power supply.

When my monitor died, I purchased a 14" VGA monitor with .28 mm dot pitch and an inexpensive VGA card. I am glad that the video port was on a card and not part of the motherboard.

With my purchase of Windows 3.0

and MS-DOS 5.0, my 640 KB of memory was insufficient. I decided to upgrade my memory beyond 640 KB and rid myself of this 128 KB memory card which had caused me problems with Prodigy, WordPerfect, and MS-DOS 5.0. I purchased an AST 286 memory card because it used standard SIMMs; but, I ran into memory parity errors after installing the card. I returned the AST card.

Options for Upgrade

My decision now was to look for another memory card, buy an upgrade kit/card to not only increase my memory but also my processor to a 386SX or greater, or to purchase a new computer.

Buying just a memory card would be the most inexpensive solution. My criteria for the card was that it use standard SIMMs so that in the future I might be able to use the memory in any new computer that I may have purchased. Zenith Data Systems memory cards do not use standard SIMMs for upgrades to 8 MHz Z-248 computers. My first try at installing a standard memory failed. If a popular standard memory card like the AST '286 memory card did not work in my Z-248, the chances of any other memory card working in my Z-248 looked slim.

I decided now was the time to purchase a 386SX. I had two options; buy a new 386SX or buy an upgrade kit/card for my Z-248 computer. First, I looked through all the past excellent Remark articles on upgrading a Z-248 computer. I also attended a Computer Conference in Washington D.C. and picked up all the handouts and literature I could find on upgrading a Z-248 computer. After much research, I decided that a SXZ-248 card (a 386SX processor) from Technology Specialists (address and phone number are at the end of this

article) in Virginia was my best bet. The ease of installation, use of standard SIMMs, price (\$494 for a 386SX, 16 MHz with 1 MB of RAM), and nearby location, made this upgrade card the clear winner. With my first half of my research complete, I embarked on my search of a new 386SX computer. Then I could compare the two options.

I searched for the cheapest IBM compatible 386SX I could find. I found a 386SX with 1 MB of RAM, 85 MB hard drive (18 ms) and a 1.4 MB (3-1/2 inch) floppy drive for \$685. The computer had the 386SX processor on the motherboard. The advantages vs the disadvantages of buying a new computer as compared to buying an upgrade card are summarized below.

Advantages

- Large hard drive (85 MB vs 60 MB)
- Faster hard drive
- Spare 3-1/2" floppy
- Newer shell with a new power supply
- Have a 286 8 MHz shell
- Have 20 MB/40 MB MFM drives

Disadvantages

- Cost \$685 vs \$494 (\$191 difference)
- New computer not upgradeable to a new processor
- Need to transfer all data to new hard drive
- Need to transfer modem, mouse, monitor, video card, and 5-1/4" drive to new computer

Looking at the above, the important advantages were a new, faster, larger hard drive and extra 3-1/2" floppy drive. The important disadvantages were the \$191 cost and the computer was not upgradeable to a new processor.

I considered the advantage of having a spare 3-1/2" floppy drive vs the disadvan-

tage of a \$191 difference to equal out. So it came down to basically getting a new 85 MB hard drive, as compared to, not being able to upgrade my processor, and the time plus troubleshooting of transferring data and hardware to a new computer.

Decision

Although not an easy decision, I decided that I have built up a lot of confidence in my Zenith Data Systems '248 computer, and would update my hard drives at a later date. I think upgrading in steps is more costly, but when a hardware part breaks, you get the latest technology for that particular hardware at that time. The other options are to fix the old part, or buy a new system whenever a major component is outdated or fails.

So I went out and bought my 386SX upgrade card. The 386SX card is priced for different configurations. There are 16 MHz, 20 MHz, and 25 MHz versions (increasing cost, of course). The standard configuration is 1 MB (80ns) installed, but you can order cards with up to 32 MB of RAM. I decided to order the 20 MHz version with 1 MB of RAM. The 20 MHz version was \$70 greater than the 16 MHz version. I decided to get just the minimum 1 MB of RAM, since extra RAM versions cost \$75 for each increment of 1 MB. I purchased 3 MB of RAM at \$42 for each 1 MB from a mail order firm and installed this RAM myself.

Installation

Installation was a snap. The card comes with an instruction booklet which I followed step-by-step. All you need is a screwdriver for the actual hardware installation. The 386SX card replaces both the Zenith Data Systems CPU and I/O cards. The 386SX contains the 386SX processor, ROM BIOS, CMOS (setup and date/time information), 1 serial, 1 parallel, and a ribbon cable to a second serial port that does not use a slot but just the opening of the slot at the back of the computer, and 8 standard SIMM memory slots where you can install 256 KB, 1 MB, or 4 MB SIMMs. SIMMs can be mixed and matched on the card in groups of 2 slots.

The first step in the instruction booklet was to access the setup on your existing Z-248 to copy down the setup information (hard drive types, floppy types, video display, etc.). I have always kept a hard copy of this information in case of battery failure. Since this one card replaces both the CPU card and the I/O card (where setup information is kept on CMOS), you must know this setup information in a later step of the installation process.

The next step is to remove the speaker jack, CPU, I/O card and any Zenith Data memory cards. Then install the 386SX card and attach the speaker jack.

The 386SX ROM BIOS supports all 46 drive types which are standard on the Z-248. Instead of CTRL-ALT-INS to get to the setup screen, the hot key is CTRL-ALT-S to get to setup on the 386SX. Since no information is in the setup (CMOS), you will get

a POST (power on self test) error. You will be prompted to press F2 to get to setup. Insert the time, date, RAM, hard drive, floppy drive, video display, etc., information on the setup screen. Save and exit the setup screen and you are ready to go.

Some ROM BIOS utilities that are available at the setup screen are a screen saver, password protection, quick boot (bypass floppy drive test and boot only from the hard disk), and soft key lock to temporarily lock the keyboard.

One feature on my Z-248 ROM BIOS which I used occasionally was the option to boot from hard drive 0, hard drive 1, floppy 0, or floppy 1. I could put different versions of autoexec.bat and config.sys on my two hard drives. This feature is not available on the ROM BIOS of the 386SX card. Beside this one minor drawback, I have been extremely pleased with my new(?) computer.

I have converted my original 8 MHz Z-248 with 512 KB of memory into a 16 MHz 80386SX with 4 MB of memory. Since I have upgraded everything else on my computer, I have a feeling that my next upgrade will be to my hard drives; but for now they are operating just fine.

Product Mentioned

SXZ-248 card (386SX processor)
Technology Specialists, Inc.

3110 Columbia Pike, Suite 303

Arlington, VA 22204

(703) 521-1886 or (800) 4-UPGRADE ✱

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record can seem rather trivial next to 20 ms, but when thousands of records are being accessed during a sort process, much more time is spent moving from track to track than actually reading data. The slow drive can easily turn a 20 second job into 65 seconds and in a day and age where computers are expected to perform tasks instantaneously, waiting for the computer to do its job is almost unacceptable.

The primary factor in determining the drive's access time is the type of head actuator mechanism used. Drives with access times less than 25 ms all have the more expensive voice coil actuators. Slower drives tend to have stepper motor actuators.

Transfer Rate

Another benchmark of drive performance is its true transfer rate, measured in Mb/second. This number is the theoretical upper limit of how much data can be transferred from the drive platters to the computer's CPU. This rate is based essentially on the speed of the drive platter rotation (usually around 3600 RPM) and the density of data on the data track. In

other words, the drive's ability to transfer data is theoretically limited to the amount of data that passes under its read/write head during a period of time. As more data is crammed onto the track, and as the platter rotates faster, then the rate will increase. In a practical sense, however, this upper rate cannot be achieved. The factors limiting performance tend to be the drive controller electronics and the computer's CPU speed. As a rule, they cannot process data quickly enough and will not perform anywhere near the theoretical rate.

There are several methods used by drive vendors that vastly improve the drive's ability to pass data to/from the CPU. One technique is to optimize the drive's interleave ratio. The concept of an interleave ratio is really very simple, the classic way of describing it is comparing it to the task of gathering your luggage off of one of the baggage conveyor belts seen in airports. Suppose you have three bags on the conveyor belt. If all three bags are situated right next to one another, you may be able to grab the first one and put it aside, but in the period to do so, the other two bags passed you by. As a consequence, you have to wait until the next pass to grab the second

bag, and possibly the third pass for the last bag. However, if the bags were placed farther apart, or if the conveyor was moving slower, then you could retrieve all three bags on the first pass. This is a simplified explanation of how the drive's head reads data from a rotating platter full of data. On slower drives with a 4:1 interleave ratio, data is read from one sector, the next three sectors are skipped, and then reading continues with the fourth sector. On faster drives, virtually all with 1:1 interleave ratios, data is read from consecutive sectors.

Another technique used to improve the drive's transfer rate is to incorporate a sector caching memory buffer on the drive controller. In some of the newer IDE drives, a 32K or 64K byte cache buffer is used which greatly improves average access times and overall data transfer rates.

Coming Next

The next article in this series will cover a variety of different drive topics, but if you don't know a IDE drive from an ESDI drive, then certainly stay tuned for Drive Lesson #2! ✱